Attention Improves Learning: A Study of Stochastic Gradient Descent (SGD)

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Abstract

We study how Stochastic Gradient Descent (SGD) behaves when training Large Language Models (LLMs) with small batches. Our main finding is that adaptive learning rates reduce the validation loss by 14.2 percent compared to a strong baseline. We also show that the improvement vanishes when the dataset is smaller than ten thousand examples.

1 Introduction

Training neural networks remains difficult because the loss surface is non-convex and noisy. Prior work (Smith 2019) claims that momentum alone is sufficient, but we find this is not true for transformer models. The NLP community has long relied on heuristics for tuning. Our contributions are threefold: a new scheduler, a theoretical bound, and an open benchmark.

2 Method

We propose the Adaptive Warmup Scheduler (AWS) which adjusts the step size every epoch. The scheduler uses a running estimate of gradient variance to decide when to decay. Experiments run on 8 GPUs with a fixed random seed of 42 and mixed precision enabled.

3 Results

AWS reaches 91.3 percent accuracy on the benchmark, outperforming the baseline by 2.1 points. However we only evaluated on English text so generalization to other languages is unknown.

4 Conclusion

Adaptive schedules matter. Future work will explore larger models and multilingual data.

References

Smith, A. (2019). Momentum is all you need. Journal of Optimization.